# Numerical Methods: Improved Euler's Method

Questions 1–14

**Question 1.** The function $y(x)$ satisfies

$$\frac{dy}{dx} = (x-1)\ln(2x+y), \qquad y(1) = 2$$

a) Use the Euler formula $y_{r+1} = y_r + h\mathrm{f}(x_r, y_r)$ with $h = 0.5$ to obtain an approximation to $y(1.5)$

b) Using your answer to part a), use the formula $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ to obtain $y(2)$ giving your answer to three decimal places.

**Question 2.** The function $y(x)$ satisfies

$$\frac{dy}{dx} = x + \ln(1+y), \qquad y(2) = 1$$

Use Euler's step by step method once, followed by using the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r),$$

where $x_{r+1} = x_r + h$ with $h = 0.1$ to obtain an approximation to $y(2.2)$ giving your answer to three decimal places.

**Question 3.** Given a second point has already been found using Euler's step by step method, use the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r),$$

where $x_{r+1} = x_r + h$ for each of the following, giving your answers to three decimal places:

a) $\frac{dy}{dx} = y \times 2^x, (0, 1), (0.75, 1.75)$ and $h = 0.75$ to find $y(1.5)$

b) $\frac{dy}{dx} = y \tan x, (1, 0.2), (1.25, 0.2779)$ and $h = 0.25$ to find $y(1.75)$

**Question 4.** Use the improved Euler's formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = h\mathrm{f}(x_r, y_r), k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ for each of the following. In each case the curve satisfies the given differential equation and passes through the given point. Give your answer to three significant figures:

a) $\frac{dy}{dx} = \ln(2x + \sqrt{y}), (1, 3)$ with $h = 2$ to find $y(3)$

b) $\frac{dy}{dx} = x - \cos^2 y, (2, 0.5)$ with $h = 0.2$ to find $y(2.2)$

c) $\frac{dy}{dx} = (3x + 2)\sqrt{x + y}, y(0) = 3$ with $h = 0.5$ to find $y(0.5)$

**Prerequisite Knowledge for Q5: Separation of Variables**

**Question 5.** A curve passes through the point $(4, e^{0.1})$ and satisfies the differential equation

$$\frac{dy}{dx} = y\sqrt{2x+1}$$

for $x > -\frac{1}{2}$

a) Use Euler's step by step method once, then the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r), \quad x_{r+1} = x_r + h$$

once, each with a step length of 0.1 to estimate the value of $y$ when $x = 4.2$
Give your answer to three decimal places.

b) Find the particular solution of the differential equation $\frac{dy}{dx} = y\sqrt{2x+1}$ in the form $\ln y = \mathrm{g}(x)$ where $\mathrm{g}(x)$ is function to be found.

c) Find the relative error of your approximation in part a) giving your answer to two significant figures.

**Prerequisite Knowledge for Q6: First Order Linear Differential Equations**

**Question 6.** A curve passes through the point $(0, 1)$ and satisfies the differential equation

$$\frac{dy}{dx} = x - \frac{1}{2}y$$

a) Use Euler's step by step method once, then the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r), \quad x_{r+1} = x_r + h$$

once, each with a step length of 0.05 to estimate the value of $y$ when $x = 0.1$

b) Find the general solution of the differential equation $\frac{dy}{dx} = x - \frac{1}{2}y$

c) Given that $y$ satisfies the differential equation in part b) and that $(0, 1)$ lies on the curve, find the value of $y$ when $x = 0.1$ giving your answer to three significant figures.

d) Comment on your accuracy in part a)

**Question 7.** A patient's drug concentration $C$ mg/L in the bloodstream changes over time ($t$ hours) according to both metabolism and a slow continuous infusion. The rate of change of concentration satisfies

$$\frac{dC}{dt} + 0.2C = 1.5e^{-0.1t}$$

Initially, the patient's drug concentration is 0 mg/L

a) Use Euler's method, with a step size of 0.5, to find an approximation for the drug concentration after 30 minutes.
b) Explain how the numerical method used in part a) could be made more accurate.
c) Refine the approximation found in part a) using $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ with a step size of 0.5. Give your answer to two significant figures.

**Question 8.** The temperature $T(°\mathrm{C})$ of a chemical solution at time $t$ minutes is modelled by

$$(15 - 0.04t^2)\frac{dT}{dt} + 0.8T = 6t$$

The initial temperature of the solution is 20°C.

a) Use Euler's method, with a step size of 0.5, to find an approximation for the temperature of the solution after 1 minute, giving your answer to three significant figures.
b) Explain how the numerical method used in part a) could be made more accurate.
c) Refine the approximation found in part a) using the formula

$$y_{n+1} = y_n + h\mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}\mathrm{f}(x_n, y_n)\right)$$

with a step size of 1. Give your answer to three significant figures.

**Question 9.** The function $y(x)$ satisfies

$$\frac{dy}{dx} = x - y$$

with the initial condition $y(0) = 1$. The fourth-order Runge-Kutta method is defined by

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = \mathrm{f}(x_n, y_n)$

$$k_2 = \mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = \mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

and $k_4 = \mathrm{f}(x_n + h, y_n + hk_3)$
Use the fourth-order Runge-Kutta method with step length $h = 0.5$ to obtain an approximation to $y(0.5)$. Give your answer to three decimal places.

**Question 10.** A curve passing through the point $(1, a)$ where $a$ is a constant satisfies

$$\frac{dy}{dx} = 5x - y$$

Applying Euler's step by step method once using $h = 0.1$, followed by the formula $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ where $x_{r+1} = x_r + h$ once from starting point $(1, a)$ gives an estimate of $y = 3.05$ when $x = 1.2$
Find the value of $a$

**Question 11.** The function $y(x)$ satisfies

$$\frac{dy}{dx} = x^2 - y$$

and the condition $y(1) = 0.5$. After applying Euler's step by step method once, it is found that $y_1 = 0.6$

a) Find the value of $h$ giving your answer to one decimal place.

b) Use the improved Euler's method, where $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ and $x_{r+1} = x_r + h$, to find $(x_2, y_2)$

**Question 12.** A curve passes through $(0, 1)$ and satisfies

$$\frac{dy}{dx} = x + y$$

Applying Euler's step by step method once, followed by $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ where $x_{r+1} = x_r + h$ once it is found that $y_2 = 1.96$

Determine the value of $h$

**Question 13.** A curve which passes through the point $(2, a)$ where $a$ is a constant, satisfies

$$\frac{dy}{dx} = 2y - 3x$$

Using $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $h = 0.5$, $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ with starting point $(2, a)$ gives an estimate of $y = 7.625$ when $x = 2.5$. Determine the value of $a$

**Question 14.** A curve passes through $(0, 1)$ and satisfies

$$\frac{dy}{dx} = 4y + 2x$$

The formula $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ is used to find that $y_1 = 2.16$ exactly. Determine the value of $h$